

Divergent viscosity paths in conjecture 00000008834

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The source claims that viscosity solution trajectories always converge to the zero set of a monotone inclusion. Neither version assumes that this zero set is nonempty. We exhibit actual divergent regularized solutions and an actual divergent viscosity flow, rather than merely pointing to an empty target set. The rate clause is not needed.

On the real Hilbert line let $A(x) = \{1\}$ for every x . This graph is monotone, since any two of its points have the same second coordinate. It is also maximal monotone: if a monotone graph extension contains (x, u) , comparison with $(x - 1, 1)$ and $(x + 1, 1)$ gives respectively $u \geq 1$ and $u \leq 1$, hence $u = 1$. Thus every monotone extension equals the original graph. Clearly A has no zero.

Actual Tikhonov solutions. For every $\varepsilon > 0$ the regularized inclusion

$$0 \in A(x) + \varepsilon x$$

is exactly $1 + \varepsilon x = 0$ and has the unique solution $x_\varepsilon = -1/\varepsilon$. Choose the genuine positive vanishing parameter

$$\varepsilon(t) = \frac{1}{t+1}, \quad t \geq 0.$$

Then its solution trajectory is $v(t) = -(t+1)$, which tends to $-\infty$ as $t \rightarrow \infty$. This parameter ranges over $(0, 1]$, so it also describes the full small-positive-parameter solution path.

Actual regularized flow. For the customary continuous viscosity dynamics

$$u'(t) + A(u(t)) + \varepsilon(t)u(t) \ni 0$$

there is an explicit solution

$$u(t) = -\frac{t+1}{2}, \quad u(0) = -\frac{1}{2}.$$

Indeed $u'(t) = -1/2$ and

$$u'(t) + 1 + \varepsilon(t)u(t) = -\frac{1}{2} + 1 - \frac{1}{2} = 0$$

for every $t \geq 0$. This smooth solution also tends to $-\infty$. In particular neither actual trajectory converges to any finite real number, in norm or weakly (the identity continuous linear functional already detects the divergence). Both common meanings of the viscosity trajectory therefore fail the unqualified convergence claim. The missing zero-existence hypothesis is essential; the example makes no claim against a theorem that includes it.

Lean correspondence. The project uses the actual real inner product and proves maximality against every monotone graph extension. It proves the full regularized inclusion and unique solution for every positive parameter, positivity and convergence to zero of $\varepsilon(t)$, actual derivatives and the regularized differential inclusion on $t \geq 0$, and actual nonconvergence of both real-time trajectories. Nonconvergence is verified via their affine integer samples and uniqueness of limits, not assumed from the absence of zeros. Lean 4.19.0 and Mathlib commit c44e0c8ee63ca166450922a373c7409c5d26b00b are pinned.